Highlights

A signal-to-drift ratio for winding-fault detectability and sensor selection in three wind-turbine generator–drive systems

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- A drift-calibrated statistic quantifies seeded-fault detectability without a model
- Phasor residuals remove the faulted-phase dependence of negative-sequence ratios
- Signatures relocate across voltage, controller and field current by machine and drive
- A decision table picks the cheapest suite with detection power 0.9 at a given extent
- Healthy referencing improves cross-system transfer in the tested machines